

Lab: Implementing Perceptual Inference

Learning Goal: Run variational message passing, compute prediction errors, and visualize belief dynamics.

Part 1: Run Message Passing

Exercise: Set up a simple 2-state model and run variational inference:

```
import numpy as np

A = np.array([[0.9, 0.1], [0.1, 0.9]]) # reliable observation
B = np.zeros((2, 2, 2))
B[:, :, 0] = np.eye(2) # action 0: stay
B[:, :, 1] = np.array([[0, 1], [1, 0]]) # action 1: switch
D = np.array([0.5, 0.5])

# Sequence of observations and actions
observations = [0, 0, 1, 1, 0] # observe state 0, then 1, then 0
actions = [0, 0, 1, 0] # stay, stay, switch, stay

qs = variational_inference(A, B, D, observations, actions)
for t, q in enumerate(qs):
    print(f"t={t}: beliefs = {q.round(3)}, obs = {observations[t]}")
```

Do the beliefs track the observations? How quickly?

Write your response here

Part 2: Prediction Error Analysis

Learning Goal: Compute and interpret prediction errors.

Exercise: Using the results from Part 1:

```
state_pe, obs_pe = compute_prediction_errors(A, B, qs, observations, actions)
for t in range(len(observations)):
    print(f"t={t}: obs_PE = {obs_pe[t].round(3)}, state_PE = {state_pe[t].round(3)}")
```

1. At which time step is the observation prediction error largest? Why?
2. Does the prediction error decrease over time? Why or why not?

Write your response here

Part 3: Precision Effects

Learning Goal: Explore how precision changes inference.

Exercise: Modify the A matrix to be less precise:

```
A_noisy = np.array([[0.6, 0.4], [0.4, 0.6]]) # unreliable observation
qs_noisy = variational_inference(A_noisy, B, D, observations, actions)

# Compare beliefs from reliable vs. unreliable observations
for t in range(len(observations)):
    print(f"t={t}: reliable = {qs[t].round(3)}, noisy = {qs_noisy[t].round(3)}")
```

How do noisier observations affect belief certainty?

Write your response here

Part 4: Free Energy Tracking

Learning Goal: Monitor free energy during inference.

Exercise: Compute free energy at each time step:

```
for t in range(len(observations)):
    if t == 0:
        prior = D
    else:
        prior = B[:, :, actions[t-1]] @ qs[t-1]
    F, acc, comp = compute_free_energy(A, qs[t], observations[t], prior)
    print(f"t={t}: F = {F:.3f}, Accuracy = {acc:.3f}, Complexity = {comp:}"))
```

Is free energy lower when observations match predictions?

Write your response here

Part 5: Reflection

In 150 words, reflect: How does watching prediction errors and free energy evolve over time deepen your understanding of perception as inference?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Running VMP	Message passing beliefs
2	Error analysis	Prediction error computation
3	Parameter effects	Precision and observation noise
4	Free energy tracking	$F = \text{Complexity} - \text{Accuracy}$
5	Reflection	Perception as inference